

ADITYA THAKARE

AI Engineer | Data Scientist | MLOps



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SUMMARY

Data Scientist with 2+ years of experience designing and deploying production-grade Generative AI systems — including RAG pipelines, multi-agent LangGraph orchestration, and end-to-end MLOps workflows. Proficient in LangGraph, LangChain, FastAPI, Docker, and MLflow. Built and deployed four production AI systems (legal AI, agentic research, predictive maintenance, financial analytics) with live production endpoints. Delivered technical training to 2,000+ professionals across 10+ industry engagements. Research background in deep learning-based ancient script recognition and applied NLP/ML.

SKILLS

Languages	Python, SQL
Generative AI	RAG · Agentic AI · LLM · LangGraph · LangChain · Gemini API · OpenAI API · Prompt Engineering · Vector Databases
Vector Stores	ChromaDB · Pinecone · HuggingFace · Gemini Embeddings · Semantic Chunking
ML / Deep Learning	Scikit-learn · XGBoost · PyTorch · TensorFlow · Keras · CNN · NLP · Transfer Learning
Backend & APIs	FastAPI · REST APIs · Streamlit · Pydantic
MLOps & DevOps	Docker · Nginx · GitHub Actions · CI/CD · MLflow · DVC · SystemD · SSL/HTTPS
Data & Visualization	Pandas · NumPy · Matplotlib · Seaborn · Power BI · Tableau

PROFESSIONAL EXPERIENCE

Data Scientist | MITU Skillologies

2023 – Present | Pune, India

- Assisted in deep learning research on **Brahmi script recognition** — conducted literature review across OCR and CNN methodologies; supported baseline implementation across four architectures (Custom Deep CNN, InceptionNet, ResNet-50, XceptionNet) on a **7,880-image, 56-class dataset**; ResNet-50 selected as optimal with **highest benchmark accuracy** across all evaluated models.
- Contributed research and implementation support to applied ML projects including **student progression prediction** (multi-source data integration, feature analysis) and **crop disease classification** — comparative experimentation across multiple model architectures.
- Designed and built **end-to-end AI systems** demonstrating LangChain RAG pipelines, neural network architectures, and MLOps workflows — serving as **primary technical SME across 10+ industry engagements reaching 2,000+ professionals** including corporate workshops and faculty development programs.
- Architected and delivered a **140-session full-stack AI curriculum** (Python through Generative AI and LangGraph) and independently led a 3-hour technical session on "Next-Gen AI: Neural Networks & Applications" for mixed technical and non-technical audiences.

PROJECTS

Conversational Legal Agent (RAG)



Live Demo | Source Code

Addresses the lack of affordable legal information access for individuals who cannot afford legal consultation.

Engineered and deployed an end-to-end conversational RAG agent specialized in Indian law. Built a complete data pipeline — web scraping, preprocessing, and semantic chunking of legal datasets. Chose **ChromaDB** over FAISS for persistent storage, enabling stateful retrieval without re-embedding. Used **Gemini text-embedding** for domain-appropriate semantic representation. Originally self-hosted on DigitalOcean with Nginx reverse proxy, SSL termination, and systemd process management; since migrated to Streamlit Community Cloud.

Tech: RAG · LangChain · ChromaDB · Gemini API · Streamlit · Nginx · Streamlit Cloud

Agentic Research System (Multi-Agent AI)

[🔗 Live Demo](#) | [⚙️ Source Code](#)

Solves shallow, single-pass AI research outputs via a self-refining multi-agent system that iterates until quality thresholds are met.

Built a production multi-agent system using **LangGraph** — chosen over LangChain's AgentExecutor for native cyclic graph support, persistent typed state, and deterministic control flow. Three specialized agents: Researcher (Tavily + Gemini synthesis), Writer (structured drafting), and Critic (quality scoring with revision trigger). **TypedDict state schema** with conditional edges loops back through Writer only when Critic score falls below threshold — preventing infinite loops and premature termination.

Tech: Agentic AI · LangGraph · LangChain · Gemini API · Tavily Search · Streamlit · Docker

Industrial Equipment Maintenance (End-to-End MLOps)

[🔗 Live Demo](#) | [⚙️ Source Code](#)

Targets the high cost of unplanned manufacturing downtime via sensor-driven predictive maintenance.

Developed a predictive maintenance classifier using **XGBoost** — selected over neural alternatives for tabular sensor data in low-data regimes. Built a reproducible **DVC** pipeline (ingest → featurize → train → evaluate) with **MLflow** experiment tracking. Fully decoupled architecture: **FastAPI** inference API deployed via **Docker** on **Render**, **Streamlit** dashboard on Streamlit Community Cloud, and **GitHub Actions** CI/CD (ruff lint + pytest on every push). Achieved **98.3% accuracy** with **91.2% F1** on imbalanced sensor data using scale_pos_weight tuning.

Tech: XGBoost · FastAPI · MLflow · DVC · Streamlit · Docker · GitHub Actions · Render

CEO Financial Dashboard (Agentic AI Backend)

Production | Client Project

Production-grade financial analytics dashboard with natural-language query capabilities for C-suite executives.

Architected an agentic AI backend using **LangGraph** orchestration with **FastAPI** serving layer. Integrated **DuckDB/Parquet** for high-performance analytical queries and **MongoDB** for metadata storage. Implemented structured outputs via **Pydantic** models with GPT-4o-mini for reliable, schema-validated natural language responses to financial queries.

Tech: LangGraph · FastAPI · DuckDB · Parquet · MongoDB · GPT-4o-mini · Pydantic

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification — DeepLearning.AI, Stanford University	Nov 2024
Generative AI, LLM & RAG — GeeksforGeeks	Dec 2025
Prompt Engineering for ChatGPT — Vanderbilt University	Mar 2024
Generative AI for Everyone — Coursera	Nov 2023
Databases and SQL for Data Science with Python — Coursera	May 2023
Docker Training Course for the Absolute Beginner — KodeKloud	Mar 2023

EDUCATION

PG Diploma in Artificial Intelligence CDAC ACTS, Pune	2023
B.Tech – Civil Engineering (CGPA: 7.78) Savitribai Phule Pune University	2021